

Absolute Value Inequalities: Special Cases and Interval Notation - Practice

Algebra II | Unit 2 | Lesson 8 Practice | TEK A2.6.F

Objective: I can write absolute value inequality solutions in interval notation, solve harder cases, and recognize the no-solution special case.

Section 1 — Recall

1. Write $|x| > 6$ in interval notation.

2. Write $|x| \leq 9$ in interval notation.

Section 2 — Apply

3. Solve $|2x - 3| < 11$, and write the answer in interval notation.

_____ ; _____

4. Solve $|x + 4| \geq 5$, and write the answer in interval notation.

_____ ; _____

Section 3 — Challenge

5. Solve $7 + |x - 2| \leq 3$.

6. Solve $3|x + 5| - 8 < 19$, and write the answer in interval notation.

_____ ; _____

Section 4 — Explain It

7. Explain why an inequality like $|x - 2| \leq -4$ has no solution at all. What is it about the isolated form that makes this happen?
